We can get partial differential of lagrange equation with respect to q:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{q}} = \mathbf{f} + \mathbf{g} = \mathbf{0}. \tag{23}$$

Thus we can get $\mathbf{f} = -\mathbf{g}$ i.e: $\mathbf{u} \odot \mathbf{v} = \mathbf{1}$

Knowing $\mathbf{P}\mathbf{1}_N = \mathbf{q} + \mathbf{d}$ and $\mathbf{P}^\top \mathbf{1}_N = \mathbf{q} - \mathbf{s} + \mathbf{d}$, we can get:

$$① \quad \mathbf{u} = \frac{\mathbf{q} + \mathbf{d}}{\mathbf{K}\mathbf{v}}, \quad \mathbf{v} = \frac{\mathbf{q} - \mathbf{s} + \mathbf{d}}{\mathbf{K}^\top \mathbf{u}} \tag{24}$$

Thus we have :

$$\frac{\mathbf{q} + \mathbf{d}}{\mathbf{K}\mathbf{v}} \odot \frac{\mathbf{q} - \mathbf{s} + \mathbf{d}}{(\mathbf{K})^\top \mathbf{u}} = \mathbf{1}$$

Then we can get the solution of q as :

$$② \quad \mathbf{q}^* = \frac{\mathbf{s}}{2} + \left((\mathbf{K}\mathbf{v}) \odot (\mathbf{K}^\top \mathbf{u}) + \frac{\mathbf{s}^2}{4} \right)^{\frac{1}{2}} - \mathbf{d}. \tag{25}$$

Knowing $\mathrm{Diag}(\mathbf{P}) = \mathbf{d}$, we can get that :

$$\mathrm{Diag}(\mathbf{P}) = \mathbf{u_i} \cdot e^{(-\mathbf{D_{ii}} + \mathbf{Diag(h_i)})/\epsilon} \cdot \mathbf{v_i} = \mathbf{d}$$

So the solution of $\mathbf{d}$ is :

$$③ \quad \mathbf{h} = \epsilon \times (\log \mathbf{d} - \log \mathbf{u} - \log \mathbf{v}) + \mathrm{Diag}(\mathbf{D})$$

From the above derivation, we can know:

$$④ \quad \mathbf{K} = e^{-\mathbf{D} + \mathbf{Diag(h)}/\epsilon}$$

By iterating through ①, ②, ③, and ④, we obtain the final Proposition

C.2 With Constrains

$$\begin{aligned} \min_{\substack{\mathbf{P} \geq \mathbf{0}, \\ \mathbf{q} \geq \mathbf{0}}} \quad & < \mathbf{D}, \mathbf{P} > -\epsilon H(\mathbf{P}) \\ \text{subject to} \quad & \mathbf{P}\mathbf{1}_N = \mathbf{q} + \mathbf{d}, \mathbf{P}^\top \mathbf{1}_N = \mathbf{q} - \mathbf{s} + \mathbf{d} \\ & \mathbf{P} \leq \mathbf{S}, \mathbf{q} \leq \mathbf{r} \end{aligned} \tag{26}$$

Adding dual variables $\mathbf{f}, \mathbf{g}, \mathbf{H}(\mathbf{H} \geq \mathbf{0})$, the lagrange equation is then:

$$\mathcal{L} = < \mathbf{D}, \mathbf{P} > -\epsilon H(\mathbf{P}) - < \mathbf{f}, \mathbf{P}\mathbf{1}_N - \mathbf{q} - \mathbf{d} > - < \mathbf{g}, \mathbf{P}^\top \mathbf{1}_N - \mathbf{q} + \mathbf{s} - \mathbf{d} > - < \mathbf{H}, \mathbf{P} - \mathbf{S} > \tag{27}$$

The partial differential of lagrange equation with respect to $\mathbf{P}$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{P_{ij}}} = \mathbf{D_{ij}} + \epsilon \log (\mathbf{P_{ij}}) - \mathbf{f_i}\mathbf{1}^\top - \mathbf{1}^\top \mathbf{g_j} - \mathbf{H_{ij}} = 0 \tag{28}$$

Thus we get :

$$\mathbf{P}_{ij} = e^{\mathbf{f_i}/\epsilon} e^{(-\mathbf{D_{ij}} + \mathbf{H_{ij}})/\epsilon} e^{\mathbf{g_j}/\epsilon} = \mathrm{Diag}(\mathbf{u})\mathbf{K}\mathrm{Diag}(\mathbf{v}) \tag{29}$$

where $\mathbf{u} = e^{\mathbf{f}/\varepsilon}$, $\mathbf{v} = e^{\mathbf{g}/\varepsilon}$, $\mathbf{K} = e^{(-\mathbf{D}+\mathbf{H})/\epsilon}$.

We can get partial differential of lagrange equation with respect to q:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{q}} = \mathbf{f} + \mathbf{g} = \mathbf{0}. \tag{30}$$

Thus we can get $\mathbf{f} = -\mathbf{g}$ i.e: $\mathbf{u} \odot \mathbf{v} = \mathbf{1}$ Similar to the Section 3.1: